

IAN ABIEL WANGSA

+65 9232 0129 | ianwangsa2@gmail.com | [linkedin.com/in/ian-abel-wangsa](https://www.linkedin.com/in/ian-abel-wangsa) | github.com/Manutd1234

EDUCATION

National University of Singapore (NUS)

Singapore

B.Sc. in Computer Science (Minor in Economics & Statistics) | GPA: 4.50/5.00

Aug 2025 – Aug 2029

- **Coursework:** Programming Methodology I & II, Data Structures & Algorithms, Linear Algebra, Discrete Structures, Calculus.

Eunoia Junior College

Singapore

GCE A-Levels (Science Stream: H2 Math, Physics, Chemistry, Economics)

Jan 2021 – Nov 2022

- **Academic Excellence:** Awarded Dean's List (2021, 2022). Received Edusave Merit and Academic Achievement Awards.
- **Varsity Softball:** 2nd Place in National School Games (2022); Awarded National Colours Award.

EXPERIENCE

Caerus Global Management

Singapore

Quantitative Summer Analyst – Apprenticeship

Aug 2026 – Present

- Architected an asynchronous pipeline using **websockets**, **aiohttp**, and **orjson** to stream high-frequency order book deltas, depth streams, and trade ticks across Binance USDS-M and Hyperliquid.
- Reconstructed stateful order books with sequence validation and built a unified 100ms cross-venue analysis grid, accounting for staleness, clock offsets, and network latency.
- Engineered microprice, order flow imbalance (OFI), and cross-venue basis features; quantified a 0.7–0.8s Binance lead-lag signal over Hyperliquid via lead/lag estimation and walk-forward cross-validation.
- Developed an event-driven backtesting engine modeling ladder-walking market impact, taker/maker fee structures, and execution delays, demonstrating signal decay and fee friction limits on HFT arbitrage.

Accenture

Singapore

Summer Intern – Application Engineer, Technology (Health & Public Services)

Apr 2026 – Jul 2026

- Engineered an enterprise **Root Cause Analysis (RCA) Generator** leveraging Agentic AI, hybrid retrieval pipelines (BGE chunking, BM25, vectorized NLP), and Azure DevOps integrations to analyze system codebases and log traces directly.
- Leveraged a custom MCP Server alongside automated Playwright authenticators to dynamically scrape SharePoint tokens, powering a Flask web frontend for seamless developer query execution.
- Architected an automated **Test Script Generator** utilizing Claude CLI Agentic AI workflow, custom sub-agents, and Python libraries; seamlessly streamlined incident response and testing workflows from multiple days to **under 30 minutes**, achieving a **95%+ boost in operational efficiency**.
- Remediated critical runtime application bottlenecks, race conditions, and production bugs across enterprise software architectures, ensuring SLA compliance and high availability.

CTES Consulting

Singapore

Software Engineer - Off-Cycle Intern

Jul 2026 – Present

- Engineered full-stack profiling and data extraction pipelines utilizing React, TypeScript, Tailwind CSS, Node.js, and PostgreSQL, deployed on AWS behind Cloudflare DNS/Security layers.
- Developed automated Web Scraping and Session Management modules using Python, Playwright, and Chromium to handle FedAuth cookie authentication and complex portal navigation.
- Built document parsing and report generation engines using **python-docx**, **pypdf**, and **openpyxl** to process candidate profiles and structured document outputs, reducing processing turnaround time from **1–2 days to a few hours**.

D+A Strategies

Remote

Quant Trainee - Apprenticeship

Mar 2026 – Present

- Design, backtest, and stress-test systematic algorithmic trading strategies by using vectorized mathematical pipelines and evaluating statistical arbitrage behaviors over unstructured financial time-series data.
- Construct quantitative modeling environments in Python to evaluate alternative pricing patterns and optimize output.

Republic of Singapore Air Force (RSAF) & Singapore Army

Singapore

Air Force Signals Commander & Specialist Cadet (NSF)

Mar 2023 – Jan 2025

- **Key Deployments:** Selected for Exercise Wallaby (Rockhampton, Australia) to establish joint SAF-RAAF radio communications; participated in SAF Day Parade (Jul 2024).
- **Unit Leadership:** Served as Comms Store IC, Gym Innovation IC, and National Education IC, leading the 160 SQN unit to win the NE CDF Award for two consecutive years (2023–2025).

PROJECTS & HACKATHONS

- BrainHack TIL-AI 2026 — DSTA Singapore** | *1st Place (Team Pandamonium)* May 2026 – Jun 2026
- **Tech Stack:** Python, Docker, FastAPI, PyTorch, NeMo, YOLOv11, BM25, Transformers, SB3
 - Built a finals-ready multimodal AI system with 5 Dockerised FastAPI microservices deployed in an air-gapped environment.
 - **ASR:** Fine-tuned NVIDIA Parakeet TDT with custom decoder, FP16 inference, sorted batching, and phrase correction.
 - **CV:** Engineered a YOLOv11 target detection pipeline for aircraft, vehicles, and vessels with threshold tuning.
 - **Noise:** Built a YOLO-surrogate PGD adversarial noise generator constrained by SSIM/RMSE budgets to fool opponent detectors.
 - **NLP:** Deployed a BM25-first RAG pipeline with alternative neural reranking layers and quantized LLM generation.
 - **AE:** Developed a MaskablePPO RL navigation agent featuring action masking, reward shaping, and fallback BFS/A* pathing.
- AlphaEngine** | *Python, Asyncio, WebSockets, Telegram Bot API, Supabase, PostgreSQL, PyArrow, Docker* Jul 2026
- Architected a real-time quant execution system using **websockets**, **aihttp**, and **orjson** to stream high-frequency L2 order book deltas, depth snapshots, and trade ticks across Binance and Hyperliquid.
 - Implemented stateful local order book reconstruction with sequence gap validation ($U/u/pu$), automated WebSocket resynchronization, and compressed Parquet storage for 100ms cross-venue grid analysis.
 - Engineered a Telegram Bot execution and monitoring add-on, broadcasting automated risk alerts, trade execution updates, and real-time PnL/position telemetry directly to remote operators.
 - Integrated a persistent Dockerized gateway server with Supabase PostgreSQL and Row Level Security (RLS) for transactional order blotter logging and strict risk limit management.
- Delta1 Trading Strategy** | *Python, Pandas, NumPy, SciPy, Statsmodels, Jupyter* Jun 2026
- Engineered high-frequency microstructure signals including Microprice, Order Flow Imbalance (OFI), and cross-venue basis z-scores; quantified a 0.7–0.8s Binance predictive lead over Hyperliquid.
 - Constructed an event-driven backtesting engine modeling full ladder-walking order book market impact, tier-based maker/taker fees, and execution latency sweeps (100ms–1000ms).
 - Validated strategy alpha via walk-forward Information Coefficient (IC) analysis, shuffled-signal control benchmarks, and round-trip cost floor decomposition to evaluate HFT arbitrage viability.

LEADERSHIP & CO-CURRICULAR ACTIVITIES

- NUS Games Development Group** | *Software Engineer (Pandamonium)* Jan 2026 – Present
- Leading technical development for *Pandamonium*, an interactive 2D murder mystery game developed using C# and Unity.
 - Designed a custom procedural board rendering pipeline from scratch and architected centralized state management systems to track game narrative progression and evidence logs.
- RunNUS & SunNUS (NUS Sports Club)** | *Executive Member* Jan 2026 – Present
- Managed event operational logistics, vendor procurement, and budget allocation for major flagship initiatives, including a flagship charity run with over 1,000 active participants.
 - Coordinated cross-functional teams to execute large-scale beach events, ensuring strict safety standards and seamless crowd control protocols.
- NUSSU VENTures** | *Operations Executive* Jan 2026 – Present
- Oversee event planning workflows, venue synchronization, and resource management for operations across campus.
 - Streamlined operational communications between campus stakeholders, venue managers, and internal student committees.
- Project BHID** | *Programmes Head* Aug 2025 – Aug 2026
- Spearheaded curriculum development tailored for adults and children with intellectual disabilities, focusing on engaging cognitive exercises and practical life skills.
 - Direct a team of student volunteers, conducting training sessions to ensure effective communication and empathetic facilitation during workshops.
- Youth Corps Singapore & Rotaract Club** | *Volunteer Educator* Aug 2025 – Jan 2026
- Taught foundational Scratch programming, logic building, and introductory Artificial Intelligence/Machine Learning concepts to primary school students.
 - Mentored young learners through hands-on coding exercises, building early interest in computational thinking and STEM.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C#, HTML, CSS
Frameworks: React, Next.js, FastAPI, Node.js, Spring Boot, Flask, Unity, Tailwind CSS, PyTorch, TensorFlow, Scikit-learn, Pandas
Developer Tools: Docker, AWS, Cloudflare, Git, Linux, Azure DevOps, Power Automate, Playwright, MediaPipe, ChromaDB
Certifications: Databricks AI/BI, AWS Cloud Practitioner (CLF-C02), Cisco Content Networking Specialist, Anthropic AI Fluency, Anthropic Claude 101, DataCamp ML Fundamentals in Python, GitHub Foundations, Docker Foundations